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Date: 11/12/2025

Submission Instructions:

Prelab:

1. No prelab

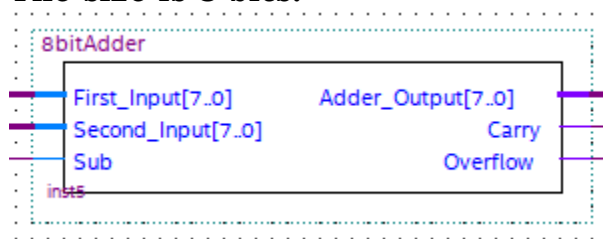
Lab:

1. Complete the scavenger hunt according to the instructions
2. Take screenshots of each component and include them in the document below (note: to receive points your NetID has to be visible in the screenshot, say in a command window that is in the background).
3. Complete this document and upload it to Canvas

Part 1: CPU Scavenger Hunt

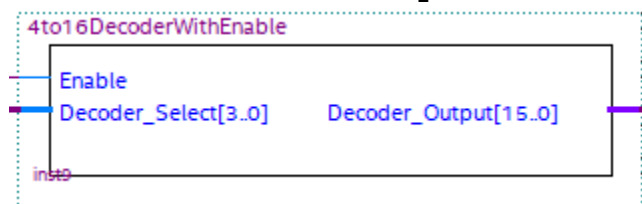
Q1. Find the **adder** inside the ALU and answer the following:

- What is the name of this component?
The name of the component is 8bitAdder.
- Is it a ripple-carry or carry lookahead adder?
The 8bitAdder is a Ripple-carry adder.
- Can it also do subtraction?
Yes, it can also do subtraction.
- What is the size of its two operands in bits?
The size is 8 bits.



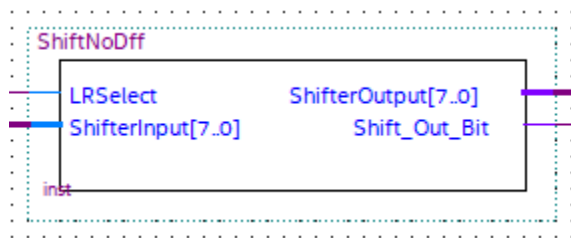
Q2. Find a 4-to-16 **decoder** and answer the following:

- What is the name of this component?
The name is 4to16DecoderWithEnable.
- Does it have an enable input?
Yes, it does have enable.
- What are the names of its outputs 4 and 6?
Decoder_OUT[4] and Decoder_OUT[6] are the names.
- Can you guess what is its function in this CPU?
To onehot encode the inputs.



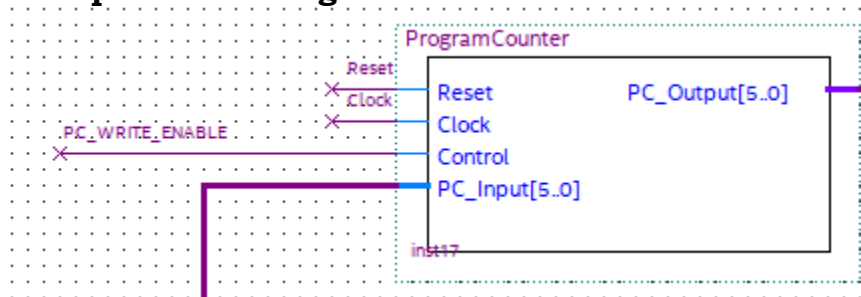
Q3. Find a **shifter circuit** and answer the following:

- What is the name of this component?
The name of the component is ShiftNoDff.
- What is the size of the input in bits?
The input size is 8 bits.
- What happens to the most significant bit on shift left?
It goes away and the second most significant bit goes there.
- What happens to the least significant bit on shift right?
It goes away and the second least significant bit goes there.



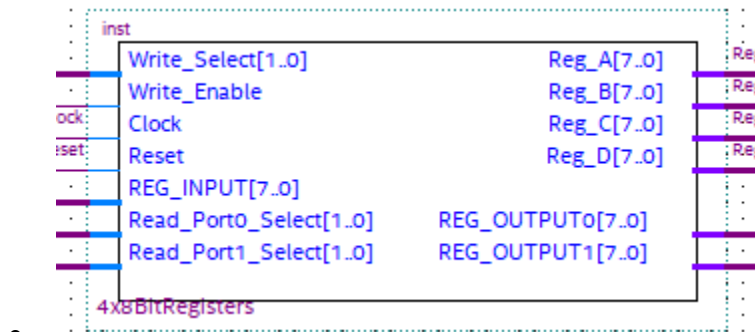
Q4. Find the **program counter** and answer the following:

- What is the name of this component?
The name of the component is ProgramCounter.
- What is the size of the output bus in bits?
The output bus is 6 bits.
- How many control lines does it have?
The program counter has one control line.
- What type of high-level circuit does it implement?
It implements a register file.



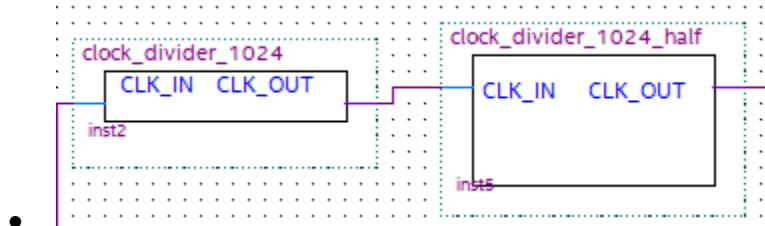
Q5. Find a **register file** with exactly 4 registers and answer the following:

- What is the name of this component?
The name of the component is 4x8BitRegisters.
- What is the size of each register in bits?
The size of the registers is 8 bits.
- What type of Flip-Flops are used to construct each register?
Each register is constructed with D Flip-flops.
- The contents of how many registers can be read at the same time?
The contents of four registers can be read at the same time.



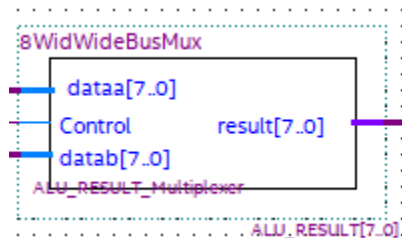
Q6. Find the two **clock dividers** for the **VideoGame_Clock** and answer the following:

- How are they implemented?
They use T Flip-flops that activates once for every time the previous one activates twice.
- They slow down the clock by a factor of X and Y. What are X and Y?
They slow it down by 1024 times and 512 times



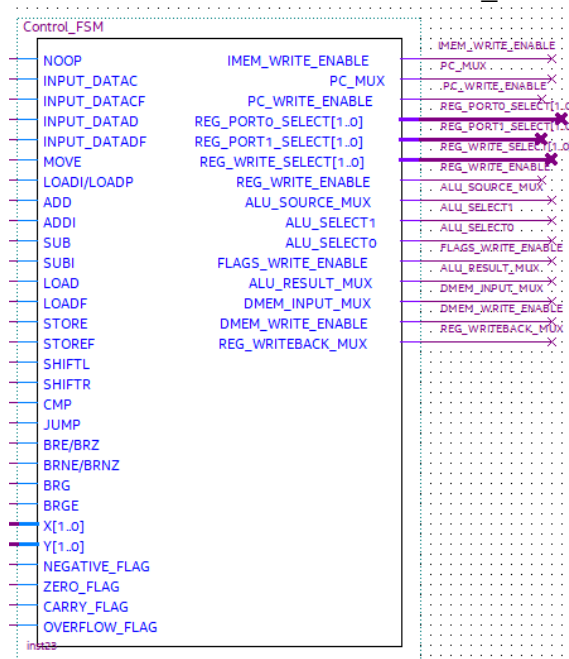
Q7. Find the **multiplexer** that sits after the ALU and takes the output of the ALU as one of its inputs. Then, answer the following:

- What is the name of this component?
The name of the component is 8WidWideBusMux.
- Where does the other input come from?
The other input comes from the IMEM.
- What is the size of each input in bits?
The size of each input is 8 bits.
- How many select lines does it have in bits?
It has one select line that is 1 bit.



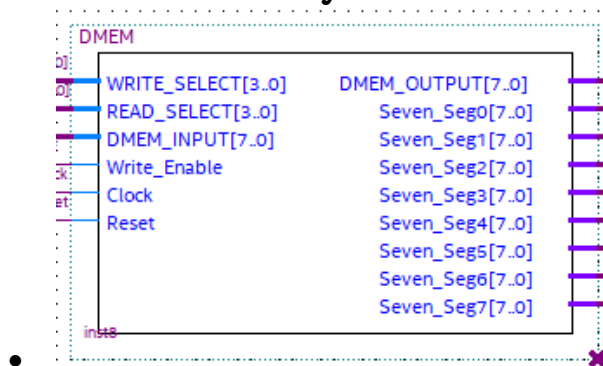
Q8. Find the circuit that outputs the signal **DMEM_WRITE_ENABLE** and then answer the following:

- In which block is this circuit located?
It is located in the i281_CPU bdf.
- What is the Boolean expression for this signal?
STOREF or STORE or INPUT_DATADF or INPUT_DATAD.



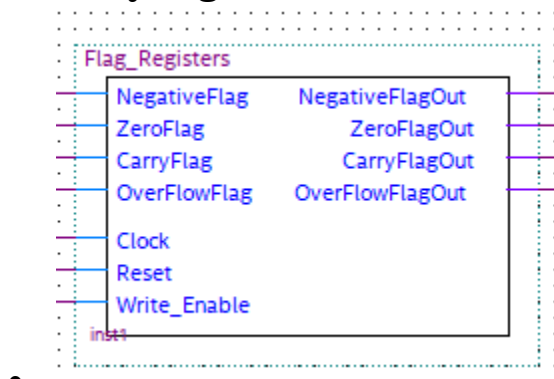
Q9. Examine the **DMEM** box and answer the following:

- What are the names of the control lines for this box?
Write_Enable and Reset.
- What the high-level component is used to store the data?
The high-level component used is Registers.
- What is the size of the data memory in bytes?
128 bits or 16 bytes.



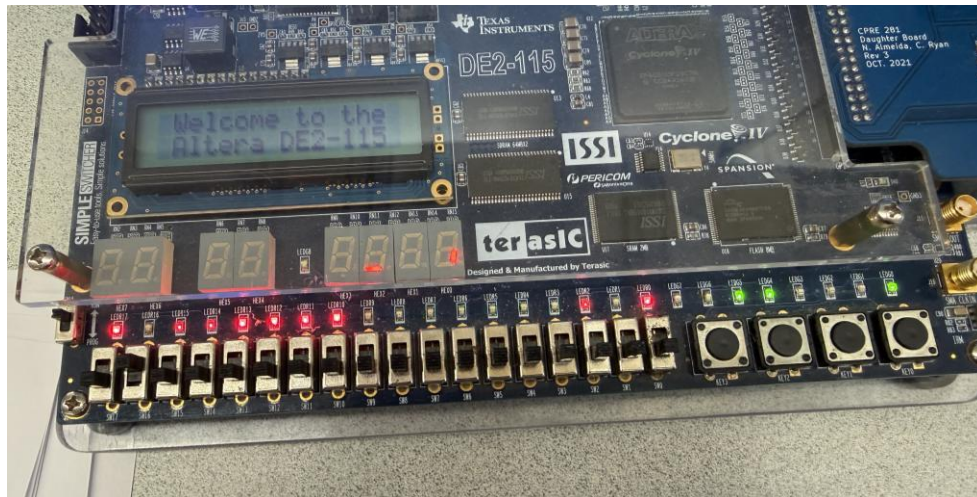
Q10. Find the **flags register** and answer the following:

- How many flags does it store?
It stores five flags.
- What are the names of these flags?
ZeroFlagOut, NegativeFlagOut, OverflowFlagOut, and CarryFlagOut



Part 2:

a) **PONG**. Take a picture of the board as you are playing the game. Selfies are OK too as long as the game is visible in the background ☺



b) **Modify the i281 Banner Program in the simulator.**

Take a screenshot of the browser window that shows the four digits of your ID on the 7-segment displays as the modified banner program is running in the simulator.

